

Habuild Data Analyst Case Study Report

Olist Brazilian E-Commerce Dataset Analysis

1. Introduction

The objective of this project was to analyze customer behavior, sales trends, payment preferences, delivery performance, and customer satisfaction using the Olist Brazilian E-Commerce Dataset.

The analysis was performed using Python and Power BI to generate business insights and identify areas where the company can improve customer experience and business growth.

2. Dataset Overview

Dataset Source

Olist Brazilian E-Commerce Dataset from Kaggle.

Tables Used

- Customers Dataset
- Orders Dataset
- Order Items Dataset
- Payments Dataset
- Reviews Dataset
- Products Dataset

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Power BI

Data Preparation Steps

- Imported CSV files using Pandas
 - Checked missing values and data types
 - Converted date columns into datetime format
 - Merged datasets using customer_id and order_id
 - Created a combined dataset for analysis
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3. Exploratory Data Analysis (EDA)

Order Trend Analysis

Monthly order trends were analyzed using purchase timestamps.

Observation

Orders increased steadily over time.

Insight

This indicates growing customer engagement and increasing platform activity.

Payment Method Analysis

Payment methods were analyzed to understand customer purchasing behavior.

Observation

Credit cards were the most commonly used payment method.

Insight

Customers prefer flexible payment options and installment-based transactions.

Review Score Analysis

Customer review scores were analyzed to evaluate satisfaction levels.

Observation

Most customers provided high review ratings.

Insight

Overall customer satisfaction appears to be positive.

Top Product Categories

Top-performing product categories were analyzed based on order volume.

Observation

A small number of product categories contributed significantly to total sales.

Insight

These categories should receive higher marketing and inventory focus.

Repeat vs One-Time Customers

Customer retention analysis was performed.

Observation

Most customers placed only one order.

Insight

The platform has strong acquisition but lower repeat purchasing behavior.

4. Hypothesis Testing

Hypothesis 1

Faster Deliveries Lead to Better Reviews

Method

Delivery time was calculated using purchase and delivery dates. Average delivery time was then compared with customer review scores.

Finding

Customers receiving faster deliveries generally gave higher ratings.

Conclusion

The hypothesis is supported.

Hypothesis 2

Customers Using Credit Cards Spend More

Method

Average payment value was compared across payment methods.

Finding

Customers using credit cards showed higher average transaction values.

Conclusion

The hypothesis is supported.

5. Key Insights

1. Orders increased consistently over time.
 2. Credit cards were the dominant payment method.
 3. Faster deliveries improved customer satisfaction.
 4. Most customers were one-time buyers.
 5. A few product categories contributed heavily to sales.
 6. Overall customer reviews were positive.
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6. Business Recommendations

Improve Customer Retention

Introduce loyalty programs, personalized offers, and customer engagement campaigns to encourage repeat purchases.

Improve Delivery Speed

Optimize logistics operations and reduce shipping delays to improve customer satisfaction.

Focus on High-Performing Categories

Increase marketing efforts and inventory planning for top-performing product categories.

Promote Credit Card Offers

Introduce cashback offers, EMI options, and bank partnerships to increase average order value.

Improve Customer Support

Monitor negative reviews closely and improve complaint resolution processes.

7. Power BI Dashboard

An interactive Power BI dashboard was created to visualize:

- Total Orders
- Total Customers
- Revenue
- Monthly Order Trends
- Payment Methods
- Review Scores
- Customer Retention Analysis
- Product Category Performance

The dashboard included KPI cards, charts, slicers, and business insights for interactive analysis.

8. Conclusion

This project helped in understanding customer behavior, sales performance, payment trends, and customer satisfaction in the Olist e-commerce dataset.

The analysis identified opportunities related to customer retention, delivery performance, and targeted marketing strategies.

Overall, this project improved my practical understanding of data analysis, visualization, and dashboard development using Python and Power BI.